

02/11/2026

Higher Categories I:

Origins

Last time Finished our discussion of generatives about toric geometry

Today Introduction to higher Categories

Origins

The bar construction. Let M be a monoid in the category of sets with multiplication $m: M \times M \rightarrow M$ and unit $u: * \rightarrow M$. We can encode the associativity and unitality axioms as a diagram

$$\begin{array}{ccccc}
 & \xrightarrow{\text{pr}_{2,3}} & & \xrightarrow{\text{pr}_1} & \\
 \xleftarrow{\text{id} \times \text{id} \times u} & & \xleftarrow{u \times \text{id}} & & \\
 & \xrightarrow{m \times \text{id}} & \xleftarrow{\text{id} \times u} & & \\
 M \times M \times M & \xleftarrow{\text{id} \times u \times \text{id}} & M \times M & \xrightarrow{m} & M & \xleftarrow{u} & * \\
 & \xrightarrow{\text{id} \times m} & \xleftarrow{\text{id} \times u} & & \\
 \xleftarrow{u \times \text{id} \times \text{id}} & & \xrightarrow{\text{pr}_2} & & \\
 & \xrightarrow{\text{pr}} & & &
 \end{array}$$

satisfying various relations that are fun to work out

Q. What is the combinatorial object indexing this?

Def. The simplex category Δ is the category of nonempty, finite, linearly ordered sets and order-preserving maps

> So every object is isomorphic to one of the form

$$[n] := \{0 < \dots < n\}$$

> There are generating maps

$$\delta^i: [n] \longrightarrow [n+1] \quad \text{"skip } i\text{"}$$

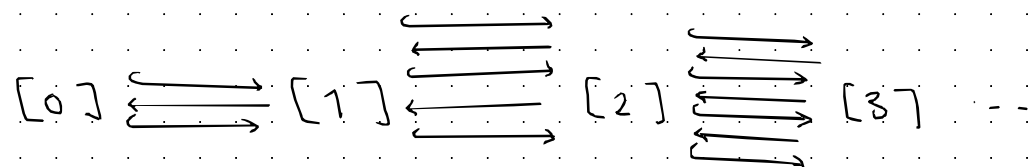
$$j \longmapsto \begin{cases} j, & j < i \\ j+1, & j \geq i \end{cases}$$

it is easy to
figure out all
relations

$$\sigma^i: [n+1] \longrightarrow [n] \quad \text{"hit } i \text{ twice"}$$

$$j \longmapsto \begin{cases} j, & j \leq i \\ j-1, & j > i \end{cases}$$

Picture



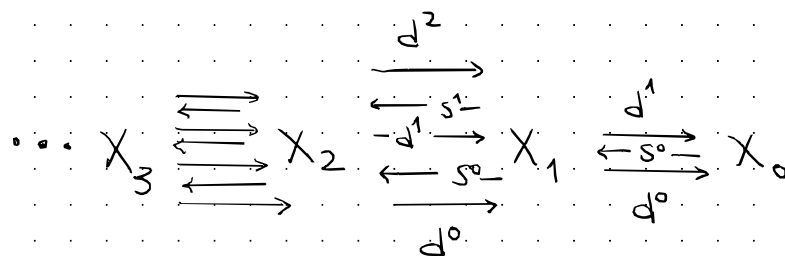
So this is dual to what we saw earlier.

Def. Let $\mathcal{C}$ be a category. The category of simplicial objects in $\mathcal{C}$ is the category $s\mathcal{C} = \text{Fun}(\Delta^{\text{op}}, \mathcal{C})$

• We often denote a simplicial object by $X_\bullet$, and call $X_n := X_\bullet([n])$ the n -simplices.

- 0-Simplices are often called vertices

- 1-Simplices are often called edges



d^i face maps

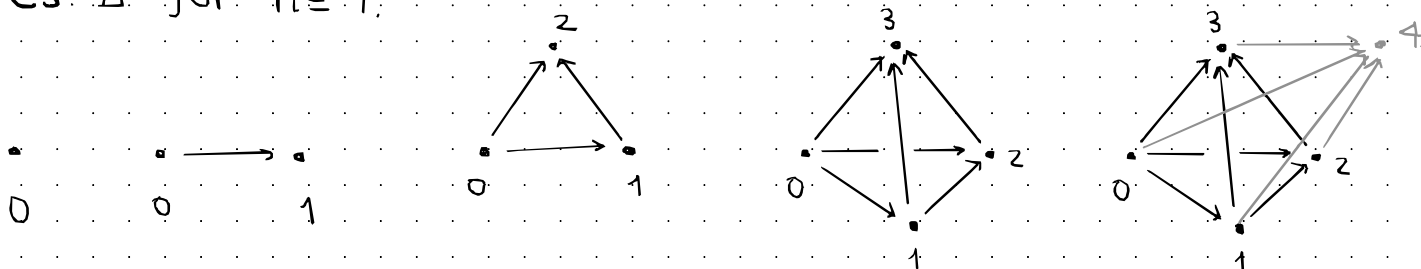
s^i degeneracy maps

Def Write Cat_1^{st} for the category of categories and functors (so this is the category of categories in the strictest possible sense) The nerve functor $N: \text{Cat}_1^{\text{st}} \rightarrow \text{sSet}$ is defined by

$$c \mapsto \left[[n] \mapsto \left\{ \begin{array}{l} \text{the set of} \\ \text{functors} \\ [n] \rightarrow c \end{array} \right\} \right]$$

Notation. For $I \in \Delta$, write $\Delta^I = N(I)$. This is also the simplicial set represented by I we write $\Delta^n = \Delta^{[n]}$

Pictures Δ^n for $n \leq 4$.

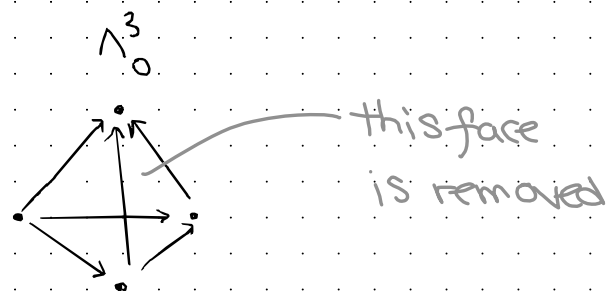
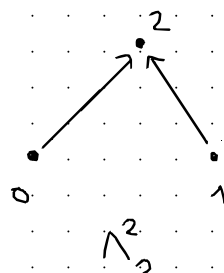
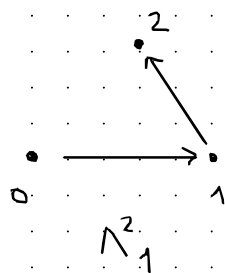
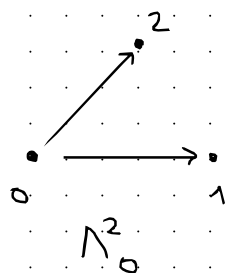


Def The k th horn in Δ^n is the simplicial subset

$$\Lambda_k^n = \bigcup_{i \in [n] \setminus \{k\}} \Delta^{[n] \setminus \{i\}}$$

face across from the
vertex k removed

Pictures



Thm (Grothendieck) The functor $N: \text{Cat}_1^{\text{st}} \rightarrow \text{sSet}$ is fully faithful. Moreover, the image consists of those simplicial sets satisfying the following equivalent conditions:

(1) Segal Condition: For each $n \geq 2$ and $t \in [n]$, the square

$$\begin{array}{ccc} X([n]) & \longrightarrow & X(\{t < \dots < n\}) \\ \downarrow & & \downarrow \\ X(\{0 < \dots < t\}) & \longrightarrow & X(\{t\}) \end{array}$$

is a pullback

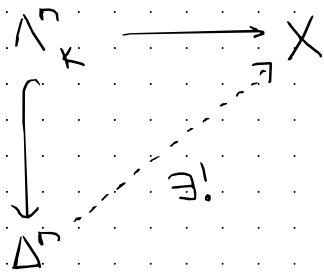
Equivalently, for each $n \geq 2$, the natural map

$$X_n = X([n]) \longrightarrow X(\{0 < 1\}) \times_{X(\{1\})} X(\{1 < 2\}) \times_{X(\{2\})} \dots \times_{X(\{n-1\})} X(\{n-1 < n\})$$

is an isomorphism.

$$\begin{array}{c} S^1 \\ X_1 \times_{X_0} X_1 \times_{X_0} \dots \times_{X_0} X_1 \end{array}$$

(1') Inner horn filling condition: For each $n > 0$ and $0 < k < n$, every diagram



has a unique filler.

Moreover, $X \in \mathbf{Set}$ is the nerve of a groupoid if and only if the following equivalent conditions hold.

(2) For each $I \in \Delta$ and partition $I = S \cup T$ such that $S \cap T = \{s\}$ is a single element, the square

$$\begin{array}{ccc} X(I) & \longrightarrow & X(S) \\ \downarrow & & \downarrow \\ X(T) & \longrightarrow & X(\{s\}) \end{array}$$

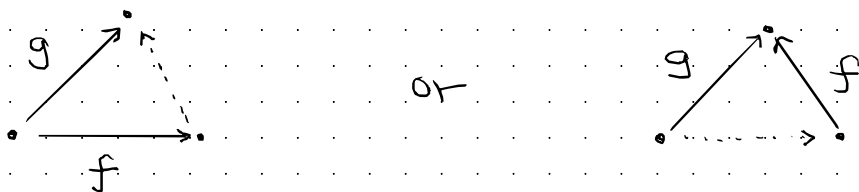
is a pullback.

(2') For each $n \geq 1$ and $0 \leq k \leq n$, every diagram

$$\begin{array}{ccc} \Lambda_k^n & \xrightarrow{\quad} & X \\ \downarrow & \nearrow \exists! & \\ \Delta_k^n & & \end{array}$$

has a unique filler

Key point This last condition says that any diagrams in the category $\mathcal{C}$ of the form



can be completed to a commutative triangle. Taking $g = \text{id}$ guarantees that f has both a left and right inverse, hence is an isomorphism.

Geometric realization

Notation. For a small category $\mathcal{C}$, write $\text{Psh}(\mathcal{C}) = \text{Fun}(\mathcal{C}^{\text{op}}, \text{Set})$ for the category of presheaves on $\mathcal{C}$ and $\mathcal{Y}: \mathcal{C} \hookrightarrow \text{Psh}(\mathcal{C})$ for the Yoneda embedding.

hiragana 'Yo' for Yoneda

Universal property of presheaves. Let $\mathcal{C}$ be a small category and $\mathcal{D}$ a category with colimits. Then precomposition with the Yoneda embedding defines an equivalence of categories

$$\text{Fun}^{\text{colim}}(\text{Psh}(\mathcal{C}), \mathcal{D}) \xrightarrow{\sim} \text{Fun}(\mathcal{C}, \mathcal{D})$$

colimit-preserving functors

$$[F: \text{Psh}(\mathcal{C}) \rightarrow \mathcal{D}] \longmapsto [\mathcal{C} \xrightarrow{\mathcal{Y}} \text{Psh}(\mathcal{C}) \xrightarrow{\bar{F}} \mathcal{D}] = F$$

The inverse is given by sending a functor $F: \mathcal{C} \rightarrow \mathcal{D}$ to its left Kan extension along the Yoneda embedding:

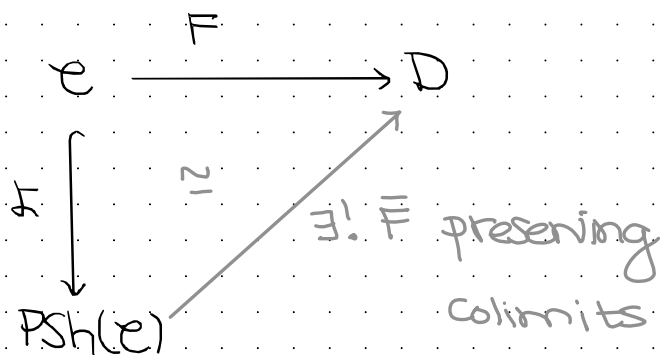
$$\bar{F}(X) := \text{Colim}_{\mathcal{Y}(c) \rightarrow X} F(c)$$

full subcategory of $\text{Psh}(\mathcal{C})_{/X}$

Moreover, every colimit-preserving functor $\bar{F}: \text{Psh}(C) \rightarrow D$ is automatically a left adjoint with right adjoint G given by the formula

$$G(d)(c) = \text{Hom}_D(F(c), d).$$

Idea, $\text{Psh}(C)$ is freely obtained from C by adjoining colimits.



Example Consider the functor

$$\begin{array}{ccc}
 \Delta & \longrightarrow & \text{Top} \\
 [m] & \longrightarrow & \Delta_{\text{top}}^m := \left\{ (t_0, \dots, t_m) \in [0, 1]^{m+1} \mid t_0 + \dots + t_m = 1 \right\} \\
 \phi \downarrow & & \downarrow \\
 [n] & \longrightarrow & \Delta_{\text{top}}^n
 \end{array}$$

$(t_0, \dots, t_m)$
 $\downarrow$
 $\left(\sum_{j \in \phi^{-1}(i)} t_j \right)_{i \in [n]}$

empty sum is zero

> This functor extends uniquely to a colimit-preserving functor

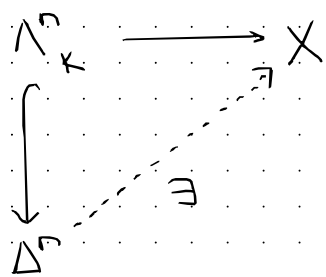
$| - | : \mathbf{sSet} \rightarrow \text{Top}$ called geometric realization

> Its right adjoint is the singular simplicial set functor

$$\text{Sing}(T)_n := \text{Hom}_{\text{Top}}(\Delta_{\text{top}}^n, T)$$

Observation. For any topological space T , the simplicial set $\text{Sing}(T)$ satisfies the following weak horn filling property.

(*) For each $n \geq 1$ and $0 \leq k \leq n$, every diagram



has a filler

Such a simplicial set is called a Kan complex or ∞ -groupoid.

Slightly tricky exercise. If $G_\bullet$ is a simplicial group, then when regarded as a simplicial set, $G_\bullet$ is a Kan complex.

The Dold - Kan Theorem

Def. Let $\mathcal{A}$ be an abelian category. Write

$$C_* : s\mathcal{A} \longrightarrow \text{Ch}_{\geq 0}(\mathcal{A})$$

for the functor

$$\left[\begin{array}{c} \cdots X_3 \begin{array}{c} \xleftarrow{\quad} \\ \xrightarrow{\quad} \\ \xleftarrow{\quad} \\ \xrightarrow{\quad} \end{array} X_2 \begin{array}{c} \xleftarrow{\quad} \\ \xrightarrow{\quad} \\ \xleftarrow{\quad} \\ \xrightarrow{\quad} \end{array} X_1 \begin{array}{c} \xleftarrow{\quad} \\ \xrightarrow{\quad} \\ \xleftarrow{\quad} \\ \xrightarrow{\quad} \end{array} X_0 \end{array} \right] \mapsto \left[\cdots \rightarrow X_2 \xrightarrow{d} X_1 \xrightarrow{d} X_0 \right]$$

$d := \sum_{i=0}^n (-1)^i d_i$

> When defining singular chains in algebraic topology, you've implicitly checked $d^2=0$ at this level of generality.

Idea. C_* is almost an equivalence of categories. But we didn't involve the s_i ! So it can't be.

Def. Write $D_*(X_*) \subset C_*(X_*)$ for the subcomplex defined by

$$D_n(X_*) := \text{im} \left(\bigoplus_{i=0}^{n-1} X_{n-1} \xrightarrow{(s^0, \dots, s^{n-1})} X_n \right)$$

The complex of normalized chains $N_*(X_*)$ is the quotient

$$N_*(X_*) := C_*(X_*) / D_*(X_*)$$

Dold-Kan Theorem. Let $\mathcal{A}$ be an abelian category

- (1) The functor $N_*: s\mathcal{A} \rightarrow \text{Ch}_{\geq 0}(\mathcal{A})$ is an equivalence of categories with explicit inverse
- (2) For each $X_\bullet \in s\mathcal{A}$, the natural map $C_*(X_\bullet) \rightarrow N_*(X_\bullet)$ is a quasi-isomorphism
- (3) For each $n \geq 0$, the diagram

$$\begin{array}{ccc} s\text{Ab} & \xrightarrow[\sim]{N_*} & \text{Ch}_{\geq 0}(\text{Ab}) \\ \downarrow \text{I-1} & & \downarrow H_n \\ \text{Top} & \xrightarrow{\pi_n} & \text{Ab} \end{array}$$

commutes. Moreover, the inverse equivalences exchange weak homotopy equivalences and quasi-isomorphisms

Implicitly, $\pi_0|X_\bullet|$ and $\pi_1|X_\bullet|$ are naturally abelian groups

- (4) N_* induces an equivalence of categories

$$s\text{Ab}[whe^{-1}] \xrightarrow{\sim} \text{Ch}_{\geq 0}(\text{Ab})[qis^{-1}] = hD_{\geq 0}(\text{Ab}).$$

Here, $\mathcal{C}[W^{-1}]$ denotes the localization of a category $\mathcal{C}$ at a collection of morphisms $W \subset \text{Mor}(\mathcal{C})$. The localization comes equipped with a functor $L: \mathcal{C} \rightarrow \mathcal{C}[W^{-1}]$ that sends every morphism in W to an isomorphism, and satisfying the following universal property:

(*) For every category D , precomposition with L defines an equivalence

$$\text{Fun}(\mathcal{C}[W^{-1}], D) \xrightarrow{\sim} \text{Fun}_{W^{-1}}(\mathcal{C}, D)$$

$\hookrightarrow$ functors that send W to isos

$$\bar{F} \longmapsto \bar{F} \circ L = F$$

